

Background:

Heat transfer between materials in an environment roughly falls into three main transfer models: radiation, conduction, and convection. Radiative transfer describes how long electromagnetic waves are emitted and absorbed by materials; Conductive transfer describes how molecules of materials transfer heat when physically in contact with one another; Convective transfer, a special form of conduction, describes how fluid flow around and between materials can transfer heat without physical contact.

Heat itself is a code word for energy flux, and in a more quantitative context heat is usually in units of energy emitted, per unit time, per unit area or $\frac{J}{m^2s}$. Given that the heat equation involves a second order differential in space and a first order differential in time, solutions can be varied in form depending on initial conditions and starting geometries. To simulate such transfers by use of a computer, discrete time steps are taken, heat is transferred at a calculated constant rate for that time step based on physical conditions, and then net heat is recalculated based on the updated values.

EnergyPlus, a DOE building energy simulation program, considers both solar radiation entering through a window, and solar radiation conducted through the face of a building. For simplicity, I will simply go over how I might simulate energy distributed through a room due to solar radiation entering the window alone.

Original Heat Equation:

$$\frac{\partial T}{\partial t} - \alpha \nabla^2 T = 0.$$

Where T is temperature, and α is a positive constant. Heat Equation with driving terms in common language:

$$\frac{\partial T}{\partial t} - \alpha \nabla^2 T = \text{radiant gain (visible surfaces)}.$$

A more useful version which includes the variability of material properties in space:

$$\rho C_p \frac{\partial T}{\partial t} - \nabla \cdot (k \nabla T) = \text{volumetric heat sources}.$$

Where k is the thermal conductivity, ρ is the mass density of the material, and C_p is the specific heat. EnergyPlus uses many contributions from different sources to calculate the heat source term, including contributions from other rooms. Given just a single room, the only outside heat contribution to any given surface is the radiation coming from the sun bouncing off other surfaces in the room. In the case of multiple rooms, heat can also transfer through the walled boundaries between rooms. Using Stefan-Boltzmann Law and some assumptions about still air within an environment, this heat source term can be found.

Radiative Transfer:

Radiative transfer should be easy to simulate in the context of the radiosity solution for visible light. Since radiosity solutions assume diffuse materials, so to must a radiation model assume diffuse heat transfer. The material in a scene can be assumed to have idealized blackbody radiator characteristics, and in general follows the Stefan-Boltzmann Law:

$$\dot{q} = \sigma T^4$$

Where $\dot{q}$ is the energy *emitted* by a surface per unit time, per unit area, σ is the Stefan-Boltzmann constant, and T^4 is the absolute temperature of the object or surface raised to the fourth power. Given that the Form Factor matrix has already been found for use in the radiosity solution, it can also be used to solve the net radiative transfer to every surface from every other surface in a scene:

$$\dot{q}_{ij} = \sigma F_{ij}(T_j^4 - T_i^4).$$

This describes the net flux *entering* surface i via surface j and the following equation describes the net flux from every surface:

$$\dot{q}_i = \sigma \sum_j^N F_{ij}(T_j^4 - T_i^4).$$

We can separate the loss term due to the temperature of an object from the gain from every other surface. For “grey bodies”, which are materials that do not perfectly re-radiate energy as black bodies do, a correction factor is added:

$$\dot{q}_i = \sigma \left[\left\{ \sum_j^N F_{ij}(\varepsilon_j T_j^4) \right\} - \varepsilon_i T_i^4 \right].$$

Where $\varepsilon < 1$. It should be noted that Stefan-Boltzmann Law is usually only valid for very high temperatures, however, EnergyPlus uses it as its approximation of radiative transfer.

Surfaces with their own emission, for example a window which has light coming through it, should be treated with an extra emission term calculated with solar direction dotted with the window surface normal.

Convective Transfer:

By far the most complicated, convective heat transfer can be approached in many ways, each of them involving their own set of assumptions and approximations. Computational Fluid Dynamics, or CFD, has been used for very simple environments to give very rigorous simulations of fluid flow in an environment. This is overkill for design applications, and so many models sum these mixing dynamics into a few simple coefficients. EnergyPlus uses an “Infrared Radiation Transfer Material” to exchange heat between zones. These surfaces absorb all incident light and have no convective coefficient associated with the air of which they are the boundary. Using this method, any transfer between the air and the walls or between the air and the air

should be calculated using the usual radiant heat exchange driving term. This is a simple modification of the equation above:

$$\dot{q}_i = \sigma \left[\left\{ \sum_j^N F_{ij} (\varepsilon_j T_j^4) \right\} + \left\{ \sum_k^M F_{ik} (\varepsilon_k T_k^4) \right\} - \varepsilon_i T_i^4 \right].$$

Where k represents the IRT surfaces. Because these theoretical surfaces would mess with lighting analysis if their form factors were calculated during the radiosity solution, their form factors need to be calculated separately, and added onto the form factor matrix calculated from the geometry of the room. In addition, if one is analyzing the heat of a given chunk of air, the only contribution to its temperature change should be net radiative loss or gain to other IRTs or to the wall of a room (the air is still and no conduction or convection occurs):

$$\Delta T_{IRT} = \frac{\Delta t}{\rho C_p} \left(\sigma \sum_m^N F_{nm} (\varepsilon_m T_m^4) - \sigma \varepsilon_n T_n^4 \right).$$

In this sum, both the IRT surfaces and the regular surface to surface radiation are included. If the air is treated as another kind of *conduction* transfer material, its adjacent walls and adjacent air must be taken as a third dimension of heat transfer, with much lower heat conduction coefficients. In order to simulate convective plumes vertically, the transfer surface at the top of each volume of air can be given a larger emissivity factor.

Breakdown into time steps:

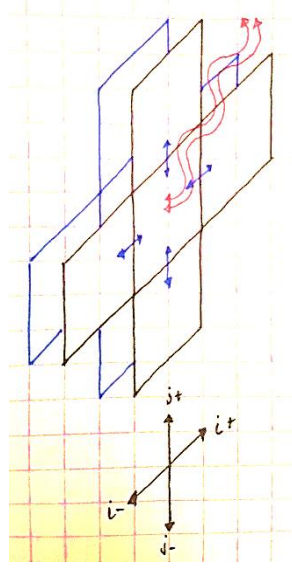
Combining the radiant gain equations with the original heat equation we started with, we can start breaking down the three-dimensional equation into easy to solve one-dimensional problems.

$$\rho C_p \frac{\partial T}{\partial t} - \nabla \cdot (k \nabla T) = \sigma \left[\left\{ \sum_l^N F_{kl} (\varepsilon_l T_l^4) \right\} - \varepsilon_k T_k^4 \right].$$

Distributing the del operator, shifting those terms to the other side, and dividing by ρC_p , we get an equation which isolates our time derivative. To perform a discrete integral, we then multiply both sides by some small Δt to calculate this time step's ΔT :

$$\Delta T = \frac{\Delta t}{\rho C_p} \left[k(\nabla^2 T) + \nabla T \nabla k + \sigma \left[\left\{ \sum_l^N F_{kl} (\varepsilon_l T_l^4) \right\} - \varepsilon_k T_k^4 \right] \right].$$

EnergyPlus uses a Finite Difference algorithm to calculate the Laplacian as a second derivative in a single direction. By finding the difference in temperature between three colinear nodes, one obtains two differences (used in the ∇T and ∇k terms), which are used to find the approximate second derivative at the center node ($\nabla^2 T$). This defines the ΔT of that surface for use in the next second derivative calculation, during the next time step. This formulation, which involves only the current set of temperatures in time, EnergyPlus calls the Adams-Moulton approach. There are many others but this one seems the most intuitive to me.



Define T_i as the center square, T_{i+1} is the next square and T_{i-1} is the previous square in the x-direction, and T_j denotes the y-direction similarly. Therefore $T_{i,j}$ is a single surface in the environment. Extend this subscript notation through the material constants ρ , C_p , and k .

Simplifying the gradient to one dimension, the spatial derivative should look something like this:

$$\nabla T = \frac{\partial T}{\partial x} \text{ OR } \frac{\partial T}{\partial y} = \frac{1}{2\Delta x} (T_{i+1} - T_{i-1}) \text{ OR } \frac{1}{2\Delta y} (T_{j+1} - T_{j-1}).$$

If the surfaces calculated are squares, $\Delta x = \Delta y$, so the second derivative should be:

$$\nabla^2 T = \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = \frac{1}{\Delta x^2} [(T_{i+1} - T_i) - (T_i - T_{i-1})] + \frac{1}{\Delta y^2} [(T_{j+1} - T_j) - (T_j - T_{j-1})].$$

To use this expanded form, the original equation is split into several parts, and then defined:

$$\Delta T = \frac{\Delta t}{\rho C_p} \left[k(\nabla^2 T) + \nabla T \nabla k + \sigma \left[\left\{ \sum_l^N F_{kl}(\varepsilon_l T_l^4) \right\} - \varepsilon_k T_k^4 \right] \right].$$

$$\Delta T = \frac{\Delta t}{\rho C_p} [M(i, j) + N(i, j) + R(i, j)].$$

Written out based on previous definitions:

$$M(i, j) = k_{i,j} \left[\frac{1}{\Delta x} [(T_{i+1,j} - T_{i,j}) - (T_{i,j} - T_{i-1,j})] + \frac{1}{\Delta x} (T_{i,j+1} - T_{i,j}) - (T_{i,j} - T_{i,j-1}) \right].$$

$$N(i, j) = \frac{1}{2\Delta x} (T_{i+1,j} - T_{i-1,j}) \frac{1}{2\Delta x} (k_{i+1,j} - k_{i-1,j}) \\ + \frac{1}{2\Delta x} (T_{i,j+1} - T_{i,j-1}) \frac{1}{2\Delta x} (k_{i,j+1} - k_{i,j-1}).$$

With some simplifications:

$$M(i, j) = k_{i,j} \left[\frac{1}{\Delta x^2} [T_{i+1,j} + T_{i-1,j} + T_{i,j+1} + T_{i,j-1} - 4T_{i,j}] \right].$$

$$N(i, j) = \frac{1}{(2\Delta x)^2} [(T_{i+1,j} - T_{i-1,j})(k_{i+1,j} - k_{i-1,j}) + (T_{i,j+1} - T_{i,j-1})(k_{i,j+1} - k_{i,j-1})].$$

$$R(i, j) = \sigma \left[\left\{ \sum_{l \neq i,j}^N F_{l \rightarrow i,j} (\epsilon_l T_l^4) \right\} - \epsilon_{i,j} T_{i,j}^4 \right].$$

Extension to larger architecture:

By introducing inner and outer surfaces as parameters in the calculations, heat can be transferred through the wall into another environment. Air to air heat transfer, simulated mostly as infrared radiation, should also be able to transfer from zone to zone through the IRT surfaces. Radiation sources such as interior lights should be treated similarly to windows, with lights like LEDs and fluorescents emitting less heat energy than incandescent lightbulbs. To simulate energy usage of a building, various EnergyPlus models can be introduced to simulate things like HVAC ventilation and moisture control. By choosing an average air temperature to maintain, optimal AC use can be calculated and added in. If ventilation is introduced, air can no longer be treated as completely still, and should be treated as a separate material with a conduction coefficient which approximates mixing air.

Computation and Storage:

Performing these calculations is easy based on the subscripts given in the equations, and adjacency of conducting surfaces is determined by their indices. Temperatures and material properties for each surface should be stored as a vector and of course the form factors are stored in a matrix. Given simple environments broken into square meshes, this method should not be terribly memory or computation intensive. However, I am not sure how computation time would scale with patch count. Since each node involves the same calculations, I would imagine that it scales roughly linearly. However, since the form factor matrix grows by the square of the number of patches, it could scale as $O(n^2)$.

Limitations:

Several limitations of this proposed calculation include:

- IRT surfaces which are not visible to one another cannot exchange energy, despite the air volumes being possibly connected.

- The approximation of air as still and uniform along large volumes is very inaccurate for ventilated buildings and will need to be revisited if HVAC systems are included in the simulation.
- Due to the approximate nature of the calculation of derivatives, the simulation is unstable given certain choices for the time step duration, square width, and conduction coefficients.
- Since the differential equation must be simulated step by step in time due to changing source terms, and since energy is accumulated over time, the computations must be done repeatedly at each step.

Speed vs EnergyPlus & Conclusion:

EnergyPlus input and output files are confusing, since the program itself runs on decades of modules being added and subtracted by dozens of engineers. It doesn't seem to take long to calculate, but this may or may not have to do with the simplicity of the models being simulated. To compare EnergyPlus calculation times to what we are capable of today, I would need to build a similarly capable simulation and run tests on the same set of models. From what I can tell, form factor matrices calculated for a radiosity solution will be much more accurate than what EnergyPlus uses, even if it takes longer. In addition, since EnergyPlus was begun long ago on old hardware, even without better computation techniques, a modern computer should be able to run EnergyPlus, as it stands, quickly.

The main reason for increasing speed of thermal load simulation is to get feedback to an architect as fast as possible to inform his/her design changes. After using EnergyPlus, with the very little experience I have, I believe that the main problem is ease of use more than speed of simulation. Given the massive amount of research one must do to even use EnergyPlus and to understand its inputs and outputs, I can see the need for outsourcing the simulation to someone else and waiting until the last possible moment to send the architecture for analysis. If EnergyPlus GUIs were easy to use, and had real visual output, I think architects would use it even if it took hours to run the simulation.

In summary, I think that reworking energy simulations for architecture using an approach informed by current radiosity solutions and "the portal method" is very possible and could be very valuable to investigate further. In my opinion, using a simulation engine to create visual feedback for designers would be even more worthwhile than creating a full DOE-approved simulator complete with moisture balance and HVAC systems. Since getting all of the precise loads is difficult, and would involve a lot of computation, the priority, the way I see it, is to get information to the architect fast so that changes to a design can be made before sending the finished product to an official simulation engine.